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Governance Quality, Public Health Spending, and Life Expectancy in South Asia: A Panel Analysis for Strategic Health Policy

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Abstract: This study evaluates whether public-sector health financing and selected governance dimensions are associated with longevity in South Asia. The analysis uses a balanced panel of 168 country-year observations for Bangladesh, India, Maldives, Nepal, Pakistan, and Sri Lanka over 1996–2023. Life expectancy at birth (LE) is modelled against domestic general government health expenditure (HE_GDP), control of corruption (CC), political stability and absence of violence/terrorism (PS), and government effectiveness (GE).Pooled ordinary least squares (OLS), feasible generalized least squares (FGLS), panel-corrected standard errors (PCSEs), and Driscoll–Kraay standard errors are reported to assess the stability of the estimates under different error structures. Diagnostic testing indicates heteroskedasticity and cross-sectional dependence, while the test for first-order serial correlation is not statistically significant. In the FGLS specification, a one-percentage-point increase in government health expenditure as a share of gross domestic product (GDP) is associated with 1.2514 additional years of life expectancy ($p < 0.01$). CC and political stability also show positive, statistically significant coefficients of 2.9629 and 2.3174, respectively. GE is negative but not statistically significant in the FGLS model (-1.3262, $p = 0.131$), and its significance is sensitive to the estimator used. Across the robustness specifications, the central pattern remains that health financing, corruption control, and political stability are important correlates of longevity. The findings support a policy approach in which budget expansion is accompanied by institutional accountability and a stable environment for implementation.

Keywords: Public health expenditure; Governance quality; Life expectancy; Strategic health policy; Corruption control; Political stability; Panel data; South Asia

1 Introduction

Life expectancy provides a broad summary of population health because it reflects cumulative exposure to disease, access to care, living conditions, and the institutional environment. Recent cross-country evidence continues to link both public health financing and institutional quality with health outcomes [1–3]. For South Asia, the policy issue is not simply how much governments spend, but whether public institutions can convert available resources into sustained improvements in survival. This study therefore examines the joint association of government health expenditure and governance quality with life expectancy in Bangladesh, India, Maldives, Nepal, Pakistan, and Sri Lanka.

The six countries have followed different health and development trajectories despite their geographic proximity and shared exposure to regional shocks. Sri Lanka and Maldives have generally recorded higher longevity, while Pakistan and Nepal have faced more persistent constraints. Such variation suggests that comparable fiscal pressure does not necessarily produce comparable health outcomes. It also motivates an explicitly institutional question: under what governance conditions is additional public health spending more likely to translate into measurable gains in life expectancy?

Health improvements in developing economies have been substantial but uneven. Global health initiatives have contributed to progress in child survival and communicable-disease control, yet the COVID-19 shock demonstrated how quickly those gains can be threatened when service delivery, coordination, and public trust are weak [4]. Recent empirical studies similarly emphasize that the effect of health spending depends on the broader policy and institutional setting rather than on expenditure alone [2, 5, 6].

This concern is especially relevant in South Asia, where fiscal space for health remains limited and the distribution of services is uneven. Additional budget allocations may have only modest effects when procurement is poorly monitored, implementation capacity is weak, or resources do not reach underserved communities. Accordingly, control of corruption (CC), political stability, and government effectiveness (GE) are treated here as institutional conditions that may help explain differences in health performance [1, 6, 7].

The pandemic also provided a recent illustration of the practical importance of governance. Administrative continuity, credible institutions, and coordinated decision-making were necessary for vaccination, emergency response, and the maintenance of routine services. Where corruption or political disruption interferes with procurement and delivery, the marginal value of public spending can be reduced. Evidence from recent international studies supports the view that governance can strengthen—or weaken—the pathway from health expenditure to health outcomes [5, 7, 8].

For this reason, the analysis does not treat governance as a background characteristic. It is examined alongside public health expenditure as a policy-relevant correlate of life expectancy. The empirical design focuses on three governance dimensions—CC, political stability and absence of violence/terrorism (PS), and GE—because they capture distinct channels through which public resources may be protected, administered, and delivered [1, 6].

Against this background, the study pursues three objectives:

1. To estimate the association between government health expenditure and life expectancy in six South Asian countries;
2. To evaluate how CC, political stability, and GE are related to life expectancy;
3. To identify institutional and policy constraints that may limit the effectiveness of public health budgets.

The contribution is threefold. First, the paper provides a South Asia-specific panel analysis covering 1996–2023, thereby extending a literature that is still dominated by global, African, or Organisation for Economic Co-operation and Development (OECD) samples. Second, it assesses health expenditure and several governance dimensions within the same empirical framework and tests the stability of the coefficients using alternative estimators. Third, the results are interpreted for strategic health policy, with particular attention to whether fiscal expansion should be accompanied by stronger accountability and political stability. This regional focus complements recent evidence from emerging and developing economies [6, 7, 9, 10].

2 Literature Review

Early work established that health outcomes are shaped by more than medical inputs. Schrecker et al. [11] emphasized the distributional consequences of globalization, while Green [4] highlighted the social determinants framework. Related studies further examined how globalization, democracy, development, and institutional conditions influence health outcomes, particularly in developing countries [12–15]. These perspectives imply that economic resources, public institutions, and the organization of policy implementation should be considered jointly when explaining population health.

Recent empirical evidence has renewed attention to this interaction. Onofrei et al. [2], studying developing members of the European Union (EU), found that public health expenditure was related to better health outcomes and that governance conditions were relevant to the effectiveness of spending. Anwar et al. [3] likewise reported a positive association between government health expenditure and life expectancy in OECD countries, while emphasizing the need for efficient use of health resources.

Evidence from developing-country panels also points to institutional moderation. Bunyaminu et al. [8] used dynamic panel methods for African countries and showed that health expenditure and GE jointly influence life expectancy. Radmehr and Adebayo [16] found that health expenditure was positively associated with life expectancy across Mediterranean countries, although the size of the effect varied across the conditional distribution of longevity.

Governance can affect health through several channels: budget integrity, administrative capacity, continuity of service delivery, and the allocation of resources across population groups. Wei et al. [9] documented links among institutional quality, government health expenditure, and human-health indicators in emerging economies. Tiwari et al. [5] used a large international panel and found that stronger governance improved maternal-health outcomes and also enhanced the translation of health expenditure into better outcomes.

The more recent evidence is not confined to one region. Kouadio and Mom [7] reported that governance quality materially shaped the health expenditure–outcome relationship in West African countries. Rahman et al. [6], using BRICS (Brazil, Russia, India, China, South Africa) data through 2023, similarly found that the contribution of health spending to Sustainable Development Goal 3 was stronger under more favorable governance conditions. These studies reinforce the argument that the same amount of public spending can produce different results depending on institutional quality.

This mechanism is consistent with earlier studies. Makuta and O'Hare [17] showed that public health spending produced larger health gains where governance was stronger, and Rajkumar and Swaroop [18] linked governance quality to the effectiveness of public expenditure. Evidence from Nigeria similarly indicated that governance affects the relationship between public health expenditure and health outcomes [19]. Hadipour et al. [1] later found that a broad institutional-quality index was associated with lower infant mortality and higher life expectancy across 158 countries. Banik et al. [20] also connected health expenditure and good governance with human development.

The direction of the spending effect is generally favorable, but its magnitude is not uniform. Sibanda et al. [21] found that institutional quality influenced how health expenditure affected under-five mortality in Sub-Saharan Africa. Ridwan et al. [22] also documented heterogeneous associations between health expenditure and life expectancy in BRICS economies. Mimi et al. [10] reported heterogeneous relationships between health expenditure and life expectancy across income groups, illustrating that structural conditions can alter the observed spending–health association. This heterogeneity supports the use of multiple estimators and cautious interpretation of coefficients.

Older contributions remain useful for identifying the underlying channels. Bokhari et al. [23] associated government health expenditure with improved outcomes in low-income settings, while Dutton et al. [24] showed that the composition of public spending can matter for health. Jack [25] cautioned that expenditure criteria and health objectives need not move together automatically. Taken together, this literature suggests that fiscal inputs should be evaluated alongside institutional quality rather than in isolation.

A regional gap nevertheless remains. South Asia has been examined less frequently as a unified panel in studies that combine government health spending with multiple governance indicators and use life expectancy as the outcome. The present study addresses that gap by analysing six countries over 28 years and by comparing pooled ordinary least squares (OLS), feasible generalized least squares (FGLS), panel-corrected standard errors (PCSEs), and Driscoll–Kraay results. The hypotheses are formulated as follows:

- Null hypothesis (H0): Government health expenditure and the selected governance indicators have no statistically significant association with life expectancy.
- Alternative hypothesis (H1): Government health expenditure and at least one of the selected governance indicators have a statistically significant association with life expectancy.

3 Data and Methodology

3.1 Data

The dataset is a balanced country-year panel covering 1996–2023 for Bangladesh, India, Maldives, Nepal, Pakistan, and Sri Lanka, yielding 168 observations. Afghanistan and Bhutan are not included because the required series are not consistently available over the full study period.

The dependent variable is life expectancy at birth (LE), expressed in years. The main fiscal variable, HE_GDP, is domestic general government health expenditure as a percentage of gross domestic product (GDP). Three governance measures are included: CC, PS, and GE. CC reflects perceptions of misuse of public authority for private benefit; PS summarizes perceived risks of political instability and politically motivated violence; and GE captures perceptions of public-service quality, civil-service performance, policy implementation, and government credibility. All series were obtained from the World Bank's World Development Indicators database. Table 1 summarizes the variables, and Figure 1 presents the country trajectories for life expectancy.

Table 1. Description of the variables

Variable	Short Form	Unit of Measurement	Source
Life expectancy at birth	LE	Years	WDI
Domestic general government health expenditure (% of GDP)	HE_GDP	% of GDP	WDI
Control of corruption	CC	Governance score	WDI
Political stability and absence of violence/terrorism	PS	Governance score	WDI
Government effectiveness	GE	Governance score	WDI

Note: WDI, World Development Indicators; GDP, gross domestic product.

Figure 1 indicates an overall rise in life expectancy across the six countries from 2000 to 2023, but the trajectories are not parallel. Maldives and Sri Lanka remain at the upper end of the regional distribution, Bangladesh and India show sustained gains, and Nepal improves from a lower base. Pakistan records the lowest levels for much of the period and comparatively slower progress. Several series also display a temporary deterioration around the COVID-19 period before recovering.

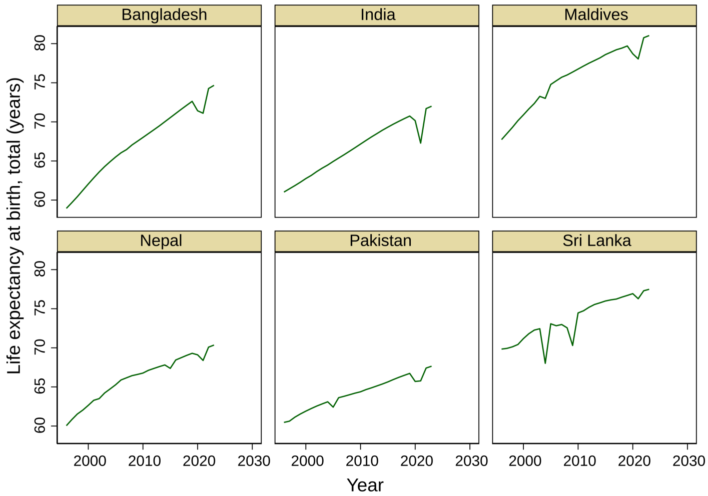

Figure 1. Trends in life expectancy at birth in six South Asian countries

The descriptive statistics in Table 2 show an average life expectancy of 68.91 years, with values between 58.94 and 81.04 years. Mean public health expenditure is 1.61% of GDP and varies considerably across countries and years. The average values of GE (-0.39), PS (-0.97), and CC (-0.64) are below zero. The dispersion in both fiscal effort and governance scores provides sufficient cross-country and intertemporal variation for examining their associations with life expectancy.

Table 2. Summary of the model

Variable	Obs.	Mean	Std. Dev.	Minimum	Maximum
LE	168	68.91393	5.323611	58.94	81.041
HE_GDP	168	1.608533	1.726423	0.1549108	9.06823
CC	168	-0.6406415	0.3325751	-1.597115	-0.0565393
PS	168	-0.9697862	0.8857168	-2.810035	1.18072
GE	168	-0.3855005	0.435416	-1.136087	0.9025494

Note: Obs., observations; Std. Dev., standard deviation; LE, life expectancy at birth; HE_GDP, domestic general government health expenditure (% of GDP); CC, control of corruption; PS, political stability and absence of violence/terrorism; GE, government effectiveness; GDP, gross domestic product.

Table 3 reports pairwise correlations. LE is most strongly correlated with HE_GDP (0.7431) and PS (0.7375), and it is also positively correlated with CC (0.4727) and GE (0.4327). Among the explanatory variables, the largest correlation is between GE and CC (0.7147). These bivariate relationships are descriptive only; the regression models are used to assess conditional associations when the explanatory variables are considered jointly.

Table 3. Correlation matrix

Variable	LE	HE_GDP	CC	PS	GE
LE	1.0000	–	–	–	–
HE_GDP	0.7431	1.0000	–	–	–
CC	0.4727	0.3484	1.0000	–	–
PS	0.7375	0.6512	0.4824	1.0000	–
GE	0.4327	0.3780	0.7147	0.5682	1.0000

Note: LE, life expectancy at birth; HE_GDP, domestic general government health expenditure (% of GDP); CC, control of corruption; PS, political stability and absence of violence/terrorism; GE, government effectiveness. A dash "–" denotes an entry omitted from the upper triangle of the symmetric correlation matrix.

3.2 Methodology

The empirical specification relates LE to HE_GDP, CC, PS, and GE for the six-country panel over 1996–2023. The coefficient on HE_GDP captures the conditional association between public health spending and longevity, while the governance coefficients measure the corresponding partial associations for corruption control, political stability, and GE.

FGLS is treated as the principal estimator because the panel has more time periods (T = 28) than cross-sectional units (N = 6) and the diagnostic results indicate heteroskedasticity and contemporaneous dependence across countries. Pooled OLS is reported as a baseline. PCSEs estimates are added to obtain inference that is robust to panel heteroskedasticity and contemporaneous correlation, and Driscoll–Kraay standard errors provide an additional check under general cross-sectional dependence and serial correlation. Using these complementary estimators allows the analysis to distinguish results that are stable from those that depend on the assumed error structure.

4 Results and Discussion

Panel stationarity is assessed with the Levin–Lin–Chu (LLC) test. LE, CC, and GE reject the unit-root null at level, whereas HE_GDP and PS do not. After first differencing, HE_GDP and PS become stationary. Table 4 reports the test statistics and p-values. The regression results are therefore interpreted with attention to these different integration properties.

Table 4. Panel unit root testing (LLC)

Variable	At Level (p-value)	At 1st Difference (p-value)	Stationary
LE	-2.5669 (0.0051)	–	Level
HE_GDP	3.6705 (0.9999)	-2.0421 (0.0206)	1st difference
CC	-2.8403 (0.0023)	–	Level
PS	-0.7084 (0.2394)	-5.1502 (<0.001)	1st difference
GE	-1.8822 (0.0299)	–	Level

Note: LLC, Levin–Lin–Chu; LE, life expectancy at birth; HE_GDP, domestic general government health expenditure (% of GDP); CC, control of corruption; PS, political stability and absence of violence/terrorism; GE, government effectiveness. A dash "–" indicates that first differencing was not required.

The pooled OLS estimates in Table 5 show positive and statistically significant coefficients for HE_GDP, CC, and PS. The HE_GDP coefficient of 1.2514 implies that, conditional on the included governance variables, a one-percentage-point increase in public health expenditure relative to GDP is associated with about 1.25 additional years of life expectancy. CC (2.9629) and PS (2.3174) are also significant at the 1% level, whereas GE is negative and statistically insignificant (p = 0.141). The positive spending result is consistent with recent evidence from EU developing countries, Mediterranean economies, and the OECD [2, 3, 16], although the size of the association is context specific.

Table 5. Pooled OLS results

Variable	Coefficient	Std. Error	t-value	p-value	Significance
HE_GDP	1.2514	0.1851	6.76	<0.001	***
CC	2.9629	1.0814	2.76	0.007	***
PS	2.3174	0.4139	5.60	<0.001	***
GE	-1.3262	0.8945	-1.48	0.141	–
_cons	71.4669	0.8122	87.99	<0.001	***

Note: OLS, ordinary least squares; HE_GDP, domestic general government health expenditure (% of GDP); CC, control of corruption; PS, political stability and absence of violence/terrorism; GE, government effectiveness; Std. Error, standard error; _cons, constant. A dash "–" indicates statistical insignificance at the reported thresholds. ***, **, and * denote significance at 1%, 5%, and 10%.

The variance inflation factor (VIF) diagnostics in Table 6 do not indicate problematic multicollinearity. The largest VIF is 2.36 for GE and the mean is 2.10, both well below commonly used screening thresholds. Accordingly, the coefficient instability observed across estimators is unlikely to be explained primarily by linear dependence among the included regressors.

The diagnostic tests in Table 7 identify two features of the error process that matter for inference. Both the Breusch–Pagan LM statistic and Pesaran's CD statistic reject cross-sectional independence ($p < 0.001$), and the heteroskedasticity test also rejects constant variance ($p < 0.001$). By contrast, the first-order autocorrelation test is not significant (p = 0.234). These findings motivate the use of FGLS and the two robust-standard-error approaches rather than reliance on pooled OLS alone.

Table 8 presents the principal FGLS estimates. HE_GDP remains positive and highly significant (1.2514; $p <$ 0.001). CC is also positive (2.9629; p = 0.005), as is PS (2.3174; $p < 0.001$). GE retains a negative sign but is not significant (-1.3262; p = 0.131). The persistence of the spending, corruption-control, and political-stability coefficients after accounting for heteroskedasticity and cross-sectional dependence strengthens the interpretation of these variables as robust correlates of longevity in the sample.

Table 6. Multicollinearity test

Variable	VIF	1/VIF
GE	2.36	0.423858
PS	2.21	0.452157
CC	2.09	0.479074
HE_GDP	1.74	0.574079
Mean VIF	2.10	–

Note: VIF, variance inflation factor; GE, government effectiveness; PS, political stability and absence of violence/terrorism; CC, control of corruption; HE_GDP, domestic general government health expenditure (% of GDP); 1/VIF, reciprocal VIF. A dash "–" indicates that 1/VIF is not applicable to the mean VIF.

Table 7. Diagnostic tests

Test	Statistic	p-value	Interpretation
VIF	2.10	–	No multicollinearity
Breusch-Pagan LM	χ^2 = 135.543	<0.001	Cross-sectional dependence exists
Pesaran's CD test	11.047	<0.001	Cross-sectional dependence exists
Heteroscedasticity	χ^2 = 89.54	<0.001	Heteroscedasticity exists
Autocorrelation	$F(1,5)$ = 1.831	0.234	No autocorrelation

Note: VIF, variance inflation factor; LM, Lagrange multiplier; CD, cross-sectional dependence; χ^2, chi-squared statistic; F, F-statistic. A dash "–" indicates that a p-value is not applicable.

Table 8. FGLS test

Variable	Coefficient	Std. Error	p-value
HE_GDP	1.2514	0.1815	<0.001
CC	2.9629	1.0607	0.005
PS	2.3174	0.4060	<0.001
GE	-1.3262	0.8774	0.131
_cons	71.4669	0.7967	<0.001

Note: FGLS, feasible generalized least squares; HE_GDP, domestic general government health expenditure (% of GDP); CC, control of corruption; PS, political stability and absence of violence/terrorism; GE, government effectiveness; Std. Error, standard error; _cons, constant.

The direction of these FGLS results is broadly consistent with recent studies that emphasize both fiscal resources and institutional quality. Kouadio and Mom [7] found that governance affects the health returns to expenditure, while Rahman et al. [6] reported stronger health-related gains from spending under better governance in BRICS economies. Tiwari et al. [5] likewise showed that governance can improve the conversion of health expenditure into health outcomes. The South Asian estimates therefore fit a wider pattern in which budgetary effort and institutional conditions operate together rather than as independent policy domains.

GE requires more cautious interpretation. Its coefficient is negative and insignificant under pooled OLS, FGLS, and Driscoll–Kraay inference, but becomes statistically significant under the PCSEs specification (p = 0.043). This estimator sensitivity differs from studies that report a consistently favorable role for GE [1, 8, 10]. In the present model, the strong correlation between GE and CC and the small number of countries may make the separate GE coefficient particularly sensitive to the covariance specification. The result should therefore not be read as evidence that GE is harmful; rather, its independent association is not robust in this sample.

The robustness estimates reported in Tables 9 and 10 support the main conclusions while also revealing where uncertainty remains. PCSEs preserve positive, significant coefficients for HE_GDP, CC, and PS, but they render GE significant and increase the PS coefficient to 2.9174. Driscoll–Kraay inference returns the same signs as the FGLS model and again treats GE as statistically insignificant. Thus, the strongest cross-estimator evidence concerns public health expenditure, CC, and political stability; the result for GE depends on the inferential method.

Table 9. PCSEs results

Variable	Coefficient	Std. Error	p-value
HE_GDP	1.2514	0.1363	<0.001
CC	2.9629	0.7737	<0.001
PS	2.9174	0.2884	<0.001
GE	-1.3262	0.6539	0.043
_cons	71.4668	0.5640	<0.001

Note: PCSEs, panel-corrected standard errors; HE_GDP, domestic general government health expenditure (% of GDP); CC, control of corruption; PS, political stability and absence of violence/terrorism; GE, government effectiveness; Std. Error, standard error; _cons, constant.

Table 10. Driscoll–Kraay standard errors test

Variable	Coefficient	Std. Error	p-value
HE_GDP	1.2514	0.3133	0.001
CC	2.9629	1.0906	0.013
PS	2.3174	0.4135	<0.001
GE	-1.3262	1.0297	0.212
_cons	71.4668	1.1198	<0.001

Note: HE_GDP, domestic general government health expenditure (% of GDP); CC, control of corruption; PS, political stability and absence of violence/terrorism; GE, government effectiveness; Std. Error, standard error; _cons, constant.

5 Conclusions and Policy Suggestions

Using a 1996–2023 panel for six South Asian countries, this study finds a stable positive association between life expectancy and three variables: domestic government health expenditure, CC, and political stability. These relationships remain statistically significant across the principal and robustness specifications. GE is less stable: its coefficient is negative in all reported models, but statistical significance appears only with PCSEs. The evidence therefore supports a focused conclusion rather than a broad claim about governance as a whole. In this sample, longevity is most consistently associated with higher public health spending, stronger corruption control, and greater political stability.

5.1 Policy Recommendations

Increase productive health investment: additional public resources should be directed toward services with broad population benefits, especially primary and preventive care, while expenditure quality is monitored alongside expenditure levels. **Strengthen anti-corruption safeguards**: transparent procurement, auditability, and clear responsibility for budget execution can reduce leakage and improve the conversion of fiscal resources into services. **Protect policy continuity**: political stability and institutional continuity are important for maintaining long-horizon health programs, supply chains, and workforce planning. **Improve administrative capacity**: although the GE coefficient is estimator-sensitive, stronger implementation systems, civil-service capability, and service management remain relevant to how budgets are executed. **Deepen regional cooperation**: The South Asian Association for Regional Cooperation (SAARC) can support exchange of comparable health-financing data, implementation practices, and preparedness strategies across South Asian health systems.

5.2 Limitations

The analysis has several limitations. It covers six countries, so the results should not be generalized mechanically to all of South Asia or to other regions. The study relies on internationally harmonized secondary indicators, which may still contain differences in measurement quality across countries and years. The estimators address important features of the error structure but do not establish causality, and reverse causation between health outcomes, fiscal choices, and governance cannot be excluded. In addition, the model includes only three governance dimensions; other institutional features, including rule of law and voice and accountability, are omitted to keep the specification parsimonious and limit collinearity.

Future work could address these limitations with research designs that are better suited to causal identification. Dynamic panel estimators such as system generalized method of moments (GMM) could model persistence in life expectancy and potential endogeneity, while lagged specifications could test whether fiscal and governance changes affect health with delay. Quantile approaches may reveal whether the associations differ between lower- and higher-longevity countries. Broader specifications could also evaluate additional governance indicators, and machine-learning methods such as random forests or the least absolute shrinkage and selection operator (LASSO) could be used for prediction or variable selection when larger panels become available.

Author Contributions

Conceptualization, M.G.K. and H.R.; methodology, H.R. and T.F.; formal analysis, M.G.K. and H.R.; writing—original draft preparation, H.R. and T.F.; writing—review and editing, M.G.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation and revision of this manuscript, the authors used ChatGPT to assist with language editing, paraphrasing, and improvements to clarity and organization. The authors reviewed and verified the final text, analyses, interpretations, and references and take full responsibility for the content of the article.

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